

Algebra II with Trigonometry

Chapter 5.3

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:

[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Solutions

1. Find the exact value of

$$\sin 255^\circ + \cos 345^\circ + \tan 225^\circ.$$

Use angle-sum identities and reference angles.

$$\sin 255^\circ = \sin(180^\circ + 75^\circ) = -\sin 75^\circ,$$

$$\cos 345^\circ = \cos(360^\circ - 15^\circ) = \cos 15^\circ,$$

$$\tan 225^\circ = \tan(180^\circ + 45^\circ) = \tan 45^\circ = 1.$$

Now compute:

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4},$$

so

$$\sin 255^\circ = -\frac{\sqrt{6} + \sqrt{2}}{4}.$$

Also,

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Hence these two terms cancel:

$$\sin 255^\circ + \cos 345^\circ = 0.$$

Therefore,

$$\sin 255^\circ + \cos 345^\circ + \tan 225^\circ = 0 + 1 = 1.$$

1

2. An angle θ satisfies $\sin \theta + \cos \theta = \frac{1}{3}$, and θ lies in Quadrant II. Find the exact value of $\tan \theta$.

Let

$$s = \sin \theta, \quad c = \cos \theta.$$

We are given

$$s + c = \frac{1}{3}.$$

Square both sides:

$$(s + c)^2 = s^2 + 2sc + c^2 = \frac{1}{9}.$$

Since $s^2 + c^2 = 1$,

$$1 + 2sc = \frac{1}{9} \quad \Rightarrow \quad 2sc = -\frac{8}{9} \quad \Rightarrow \quad sc = -\frac{4}{9}.$$

Now s and c are roots of

$$t^2 - \frac{1}{3}t - \frac{4}{9} = 0.$$

Multiply by 9:

$$9t^2 - 3t - 4 = 0.$$

Thus

$$t = \frac{3 \pm \sqrt{9 + 144}}{18} = \frac{3 \pm 3\sqrt{17}}{18} = \frac{1 \pm \sqrt{17}}{6}.$$

In Quadrant II, $\sin \theta > 0$ and $\cos \theta < 0$. Hence

$$\sin \theta = \frac{1 + \sqrt{17}}{6}, \quad \cos \theta = \frac{1 - \sqrt{17}}{6}.$$

Therefore

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{1 + \sqrt{17}}{1 - \sqrt{17}}.$$

Rationalizing,

$$\tan \theta = \frac{(1 + \sqrt{17})^2}{1 - 17} = \frac{18 + 2\sqrt{17}}{-16} = -\frac{9 + \sqrt{17}}{8}.$$

$$\boxed{-\frac{9 + \sqrt{17}}{8}}$$

3. Without using a calculator, determine the sign and exact value of

$$\sec\left(-\frac{7\pi}{6}\right) + \csc\left(\frac{11\pi}{6}\right).$$

First simplify each term.

Since cosine is even,

$$\cos\left(-\frac{7\pi}{6}\right) = \cos\left(\frac{7\pi}{6}\right).$$

Now

$$\frac{7\pi}{6} = \pi + \frac{\pi}{6},$$

so

$$\cos\left(\frac{7\pi}{6}\right) = -\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}.$$

Thus

$$\sec\left(-\frac{7\pi}{6}\right) = \frac{1}{-\sqrt{3}/2} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}.$$

Next,

$$\frac{11\pi}{6} = 2\pi - \frac{\pi}{6},$$

so

$$\sin\left(\frac{11\pi}{6}\right) = -\sin\frac{\pi}{6} = -\frac{1}{2}.$$

Hence

$$\csc\left(\frac{11\pi}{6}\right) = \frac{1}{-1/2} = -2.$$

Therefore

$$\sec\left(-\frac{7\pi}{6}\right) + \csc\left(\frac{11\pi}{6}\right) = -\frac{2\sqrt{3}}{3} - 2.$$

This is clearly negative.

$$\boxed{-2 - \frac{2\sqrt{3}}{3}}$$

and the sign is negative.

4. A point $P(x, y)$ lies on the terminal side of an angle θ in Quadrant III, and

$$x + y = -7, \quad \frac{x}{y} = \cot \theta = \frac{3}{4}.$$

Find the exact values of $\sin \theta$, $\cos \theta$, $\tan \theta$.

We are given

$$\frac{x}{y} = \frac{3}{4}.$$

So $x : y = 3 : 4$. Since the point is in Quadrant III, both x and y are negative. Write

$$x = -3k, \quad y = -4k$$

for some positive k .

Using $x + y = -7$,

$$-3k - 4k = -7 \Rightarrow -7k = -7 \Rightarrow k = 1.$$

Hence

$$x = -3, \quad y = -4.$$

Now

$$r = \sqrt{x^2 + y^2} = \sqrt{9 + 16} = 5.$$

Therefore

$$\sin \theta = \frac{y}{r} = -\frac{4}{5}, \quad \cos \theta = \frac{x}{r} = -\frac{3}{5}, \quad \tan \theta = \frac{y}{x} = \frac{-4}{-3} = \frac{4}{3}.$$

$$\boxed{\sin \theta = -\frac{4}{5}, \quad \cos \theta = -\frac{3}{5}, \quad \tan \theta = \frac{4}{3}}$$

5. Two sides of a triangle have lengths 13 and 20, and the included angle is 150° . Find the area of the triangle.

Use the formula

$$\text{Area} = \frac{1}{2}ab \sin C.$$

Here $a = 13$, $b = 20$, and $C = 150^\circ$. Thus

$$\text{Area} = \frac{1}{2}(13)(20) \sin 150^\circ.$$

Since

$$\sin 150^\circ = \sin 30^\circ = \frac{1}{2},$$

we get

$$\text{Area} = 130 \cdot \frac{1}{2} = 65.$$

65

6. An observer is 5280 ft from the launch pad of a rocket. At one moment the angle of elevation is 45° ; one minute later it is 60° . Assuming the rocket rises straight up at constant speed during that minute, how many feet does it rise?

Let the horizontal distance from observer to launch pad be

$$d = 5280 \text{ ft.}$$

If the rocket's height is h , then

$$\tan \theta = \frac{h}{d}.$$

At the first moment, $\theta = 45^\circ$:

$$h_1 = 5280 \tan 45^\circ = 5280.$$

One minute later, $\theta = 60^\circ$:

$$h_2 = 5280 \tan 60^\circ = 5280\sqrt{3}.$$

So the rise during that minute is

$$h_2 - h_1 = 5280(\sqrt{3} - 1).$$

$$5280(\sqrt{3} - 1) \text{ ft}$$